

INVESTIGATION REPORT

Issue 1006: Duplicate hardware-loop jump guard

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1006
Disposition	Waiver, with a stated software rule
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1006
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	COREV_PULP=1, FPU=0 (RTL master 6033d2b)
Disposition	Waiver, with a stated software rule
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	10 cited, 8 present, 2 not retained

Disposition meaning. The code is unreachable only while software obeys an explicitly stated rule. The rule is part of the disposition and is recorded in the Action section.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

cv32e40p_controller.sv:632 and :642 are the 'if (~jump_done_q)' guards inside the DECODE default arm when the instruction in ID is the last of an active HWLoop0 (line 630) or HWLoop1 (line 640). The uncovered AllFalse arms need jump_done_q==1 there. jump_done_q is set only when a jump was issued while ID was stalled (jump_done & ~id_ready_i) and is re-evaluated in DECODE only if the FSM stays in DECODE, which at line 627 requires hwlp_end0_eq_pc_plus4 || hwlp_end1_eq_pc_plus4, i.e. the OTHER loop's end register equals this loop's end + 4. Hardware alone reaches both arms: the cover engine hits line 632 and 642 AllFalse in 9 cycles, in both cases with end1 == end0 + 4 (illegally nested loop registers, plus 32-bit address wrap). That register layout violates the PULP HWLoop nesting rule (outer end >= inner end + 8, the same invariant as #1009/#1005). Under that rule the arms are unreachable: z3 proves it (unsat with the invariant, sat without, model end0=pc+4, end1=pc+8), and a k-induction proof (k=12) of the exact guard passes for both lines with the single assumption that the two end registers never differ by exactly 4 bytes (abc pdr on the same properties did not converge in 2.4 h, kept as negative result). Behaviour if a program does violate the rule: the guard just suppresses a second pc_set/hwlp_dec_cnt for the same jump, benign. Disposition: waiver with the stated register-content invariant, now backed by a formal proof rather than the Cadence SLEC equivalence argument.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_controller.sv:632 and :642 'if (~jump_done_q)' AllFalse arms: reachable only when the two HWLoop end registers differ by exactly one 32-bit instruction ($\text{end1} == \text{end0} + 4$ or $\text{end0} == \text{end1} + 4$), which the PULP HWLoop nesting rule (outer end at least two instructions past inner end) forbids. Under the invariant $\text{hwlp_end_addr}[1] != \text{hwlp_end_addr}[0] + 4$ && $\text{hwlp_end_addr}[0] != \text{hwlp_end_addr}[1] + 4$ both arms are formally unreachable (k-induction $k=12$, runs/v1-cv32e40p-1006/formal/p_line632_inv_kind and p_line642_inv_kind; z3 runs/v1-cv32e40p-1006/smt). Hardware does not enforce the rule: bounded cover reaches both arms in 9 cycles with an illegal register set (runs/v1-cv32e40p-1006/formal/cover). Waive both arms as software-illegal with that invariant; the guarded code only suppresses a duplicate jump and is benign if violated.

Checker used

checker_1006.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/cover/sby.log	present	cover PASS (153.8 s): both taken arms at step 7, jump_done_q in DECODE at step 8, both AllFalse (hole) arms at step 9 -> reachable by hardware with illegal loop registers
E2	counterexample	formal/cover/job/engine_0/trace4.vcd	not kept	line 632 hole witness: loop0=[pc, pc+4] cnt0=0xffe, end1=pc+8 (=end0+4), id_ready=0 -> jump_done_q=1 next cycle, FSM stays DECODE (cover_hole_line632_hw_only.png)
E3	counterexample	formal/cover/job/engine_0/trace3.vcd	not kept	line 642 hole witness: mirror case, end0=end1+4 (cover_hole_line642_hw_only.png)
E4	smt-model	smt/line632_with_invariant.out	present	z3 unsat: stay-in-DECODE condition + line-632 guard impossible under $\text{end1} \geq \text{end0} + 8$
E5	smt-model	smt/line632_without_invariant.out	present	z3 sat without the rule: $\text{end0} = \text{pc} + 4$, $\text{end1} = \text{pc} + 8$
E6	smt-model	smt/line642_with_invariant.out	present	z3 unsat (line 642 mirror)
E7	proof	formal/p_line632_inv_kind/sby.log	present	k-induction (smtbmc yices, $k=12$) PASS (1057 s) under assume($\text{end1} != \text{end0} + 4$ && $\text{end0} != \text{end1} + 4$): line 632 AllFalse unreachable
E8	proof	formal/p_line642_inv_kind/sby.log	present	k-induction ($k=12$) PASS (1056 s) under the same assumption: line 642 AllFalse unreachable
E9	proof	formal/p_line632/sby.log	present	negative result kept: unconstrained abc pdr TIMEOUT 30 min (property is false without the invariant; cover found the witness first)
E10	proof	formal/p_line632_inv/sby.log	present	negative result kept: abc pdr with invariant did not converge in 2.4 h; k-induction used instead

Paths are relative to runs/v1-cv32e40p-1006. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- smt-model. A satisfying assignment from the SMT solver for the stated query.

Artefact(s) E2, E3 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

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Limitations and scope

- The waiver holds only while software obeys the rule stated in Action. If software can violate that rule, the waiver does not hold and the code is reachable.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

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Provenance and measurement

Run directory	runs/v1-cv32e40p-1006
Title source	Title and configuration read back from reports/issues/cv32e40p_issue_1006.pdf, the previously published report for this issue.
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024 (issue body): waived in v2 after Cadence JasperGold SLEC showed removing lines 632/638/642/648 leaves top-level outputs unchanged (unreachable)
Assessed confidence	0.8. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	2

Measured tool and step time

Step	Measured
read issue + locate lines 632/642 + jump_done_q analysis	7.0 min (420 s)
v1 asserts + 5 covers	5.0 min (300 s)
cover: holes reached, read witnesses	2.6 min (154 s)
z3 with/without invariant	60.0 s
v2 asserts with invariant, pdr (no convergence)	10.0 min (600 s)
k-induction k=12 both lines	17.6 min (1057 s)
report	7.0 min (420 s)
Total	50.2 min (3011 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

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